Types of Classification Error

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Types of Classification Error before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rows Columns and Data Types

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Types of Classification Error trace and record the first step you can explain without help.

Use W2: Explain the two error types for Confusion matrices. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Types of Classification Error explanation against the supplied facts before discussing it.

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Count a fresh evaluation independently

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Types of Classification Error target to a trace and a justified conclusion.

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Types of Classification Error case and show the deciding evidence.

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Positive means sports. A sports book predicted other and a non-sports book predicted sports occur. Name each error and its consequence for the sports shelf.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 13 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Types of Classification Error adds to Transparent Book Recommender and keep the evidence.

Separate recommending an unsuitable book from failing to surface a suitable one.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Build the count matrix

> Build the count matrix. Apply the supplied Types of Classification Error facts and keep your reasoning.

Use all eight worked IDs. Draw a matrix with reference rows and **prediction** columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Explain the two error types

> Explain the two error types. Apply the supplied Types of Classification Error facts and keep your reasoning.

Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit totals and equal accuracy

Audit totals and equal accuracy. Apply the supplied Types of Classification Error facts and keep your reasoning.

Give the worked row totals, column totals, total count and accuracy. Compare its errors with [[3,0],[3,2]] using the same axes and eight reference records.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Count a fresh evaluation independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Count a fresh evaluation independently. Apply the supplied Types of Classification Error facts and keep your reasoning.

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My reasoning and evidence

W5 Interpret a zero count carefully

> Interpret a zero count carefully. Apply the supplied Types of Classification Error facts and keep your reasoning.

A report gives zero **false positive**s but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Types of Classification Error

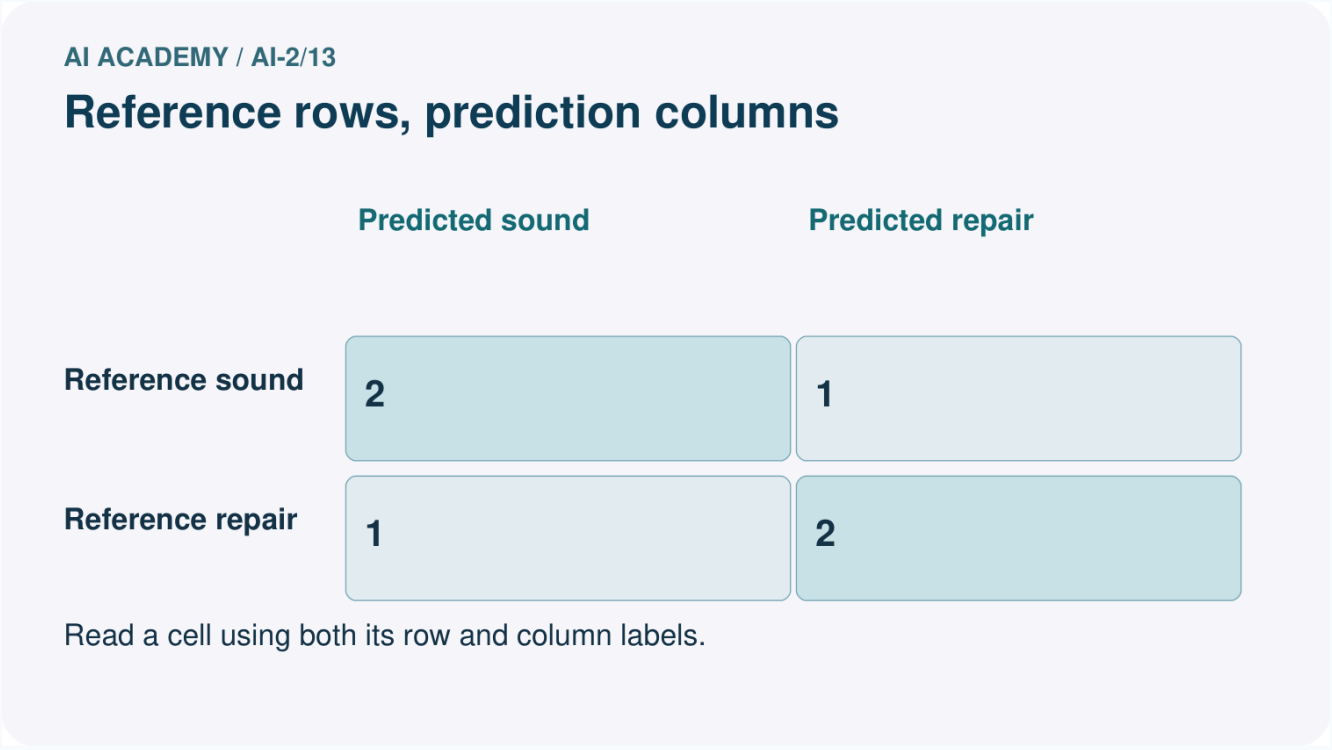
> A fictional book-condition checker predicts repair or sound. Explain the labelled step before

> checking the changed condition.

A fictional book-condition checker predicts repair or sound.

Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/13

Types of Classification Error

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare axes

Rows: reference sound,repair. Columns: predicted sound,repair.

2. Fill sound row

Two true negatives and one false positive: [2,1].

3. Fill repair row

One false negative and two true positives: [1,2].

4. Read result

Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

CHANGE ONE THING

Change one repair/sound prediction to repair/repair after a real model revision.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which error causes a sound book to be sent for unnecessary repair review?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-condition checker predicts repair or sound.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

> 1. Declare axes

2. Read result

3. Fill repair row

4. Fill sound row

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Read result Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Each case enters one cell according to its reference and predicted label.

The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table’s interpretation even if its diagonal sum remains unchanged.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change one repair/sound **prediction** to repair/repair after a real model revision. Which conclusion is supported by the supplied facts?

1. Repair row becomes [0,3]; agreement becomes 5/6.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair row becomes [0,3]; agreement becomes 5/6. One **false negative** moves to **true positive**; the sound row stays [2,1].

Repair a missing building block

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

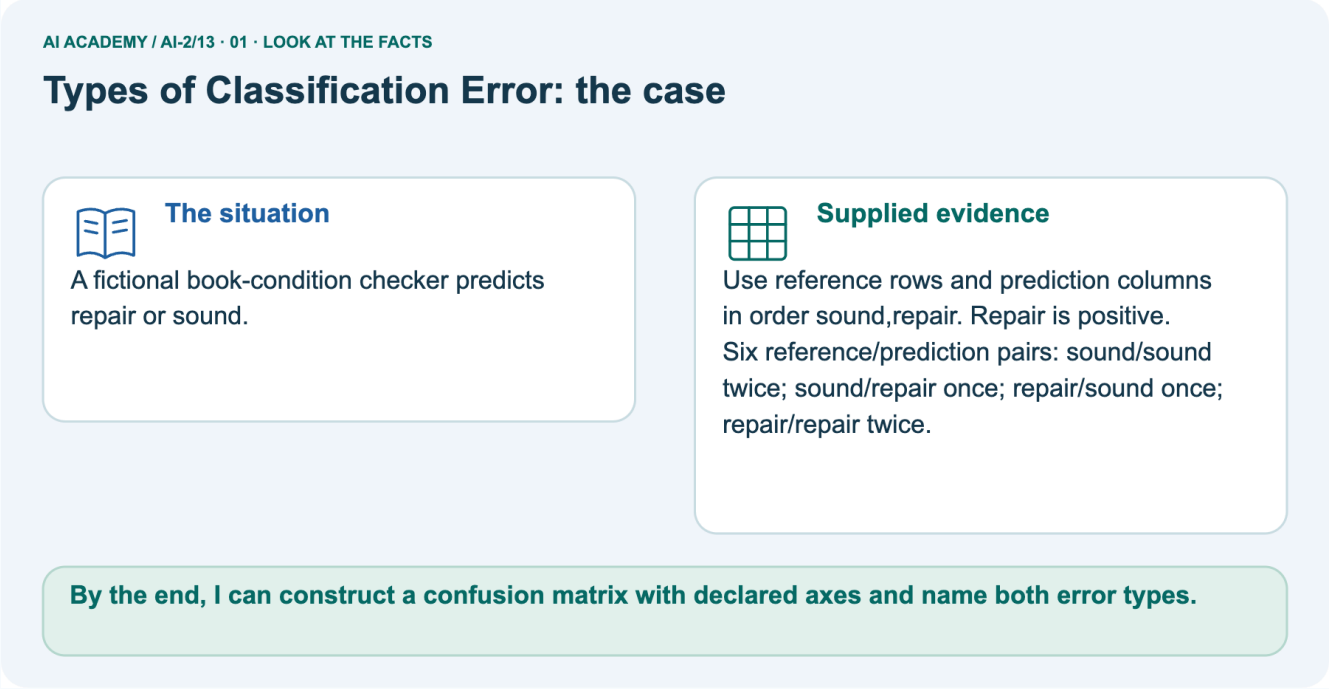
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Types of Classification Error picture from its labels, then record a relevant limit.

01. Types of Classification Error: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

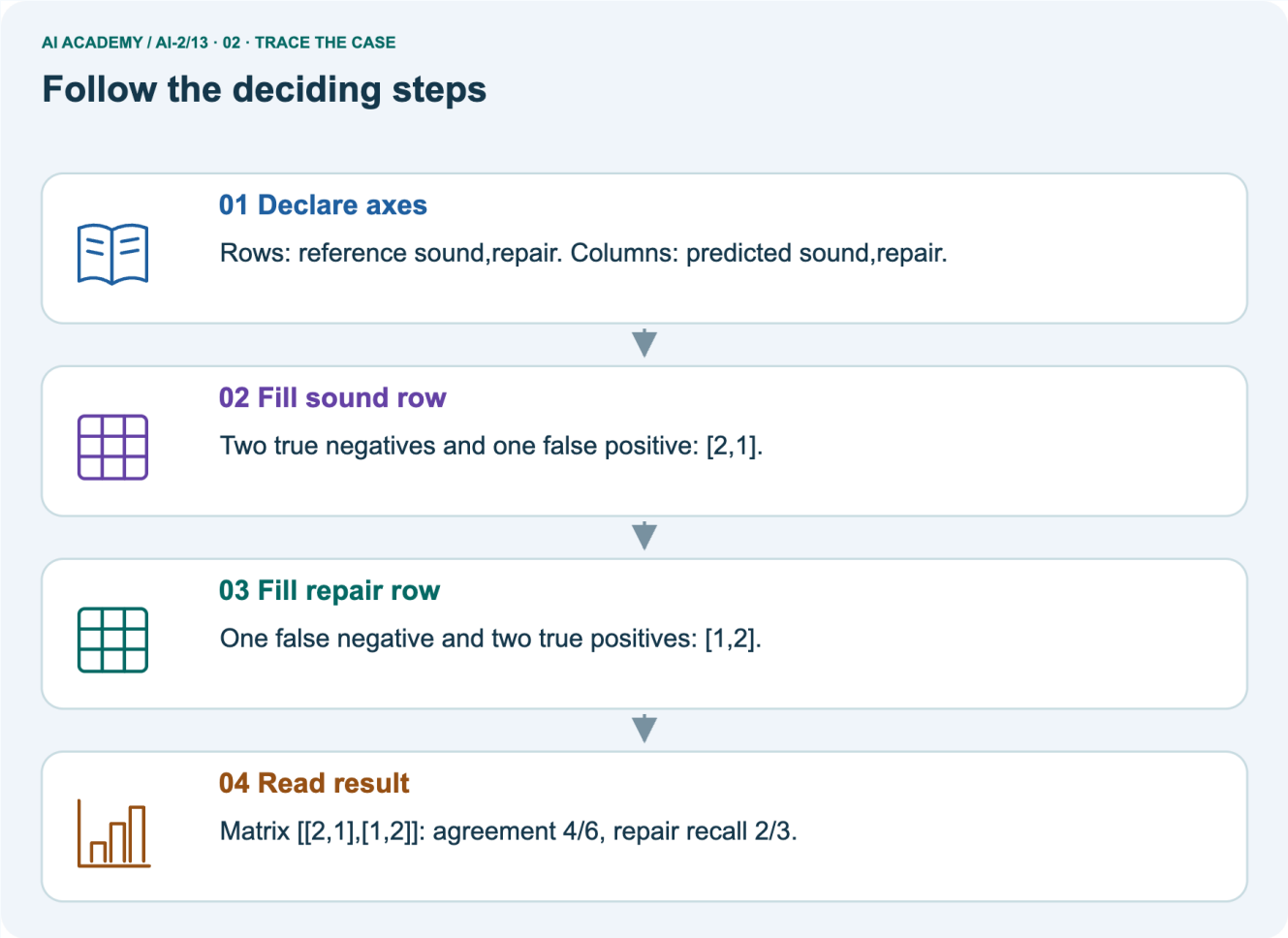
A fictional book-condition checker predicts repair or sound. Supplied facts: Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

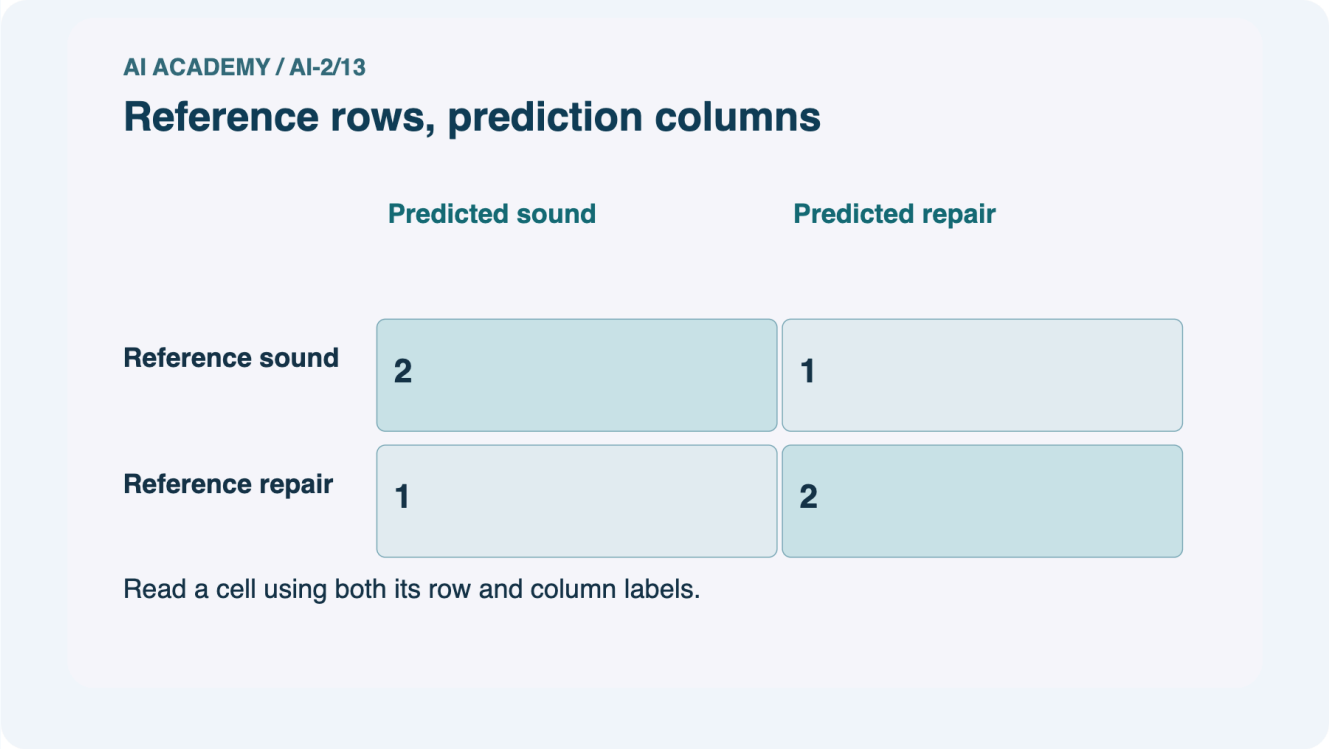
Declare axes: Rows: reference sound,repair.

Columns: predicted sound,repair. → Fill sound row: Two **true negative**s and one **false positive**: [2,1]. → Fill repair row: One **false negative** and two **true positive**s: [1,2]. → Read result: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Reference rows, prediction columns



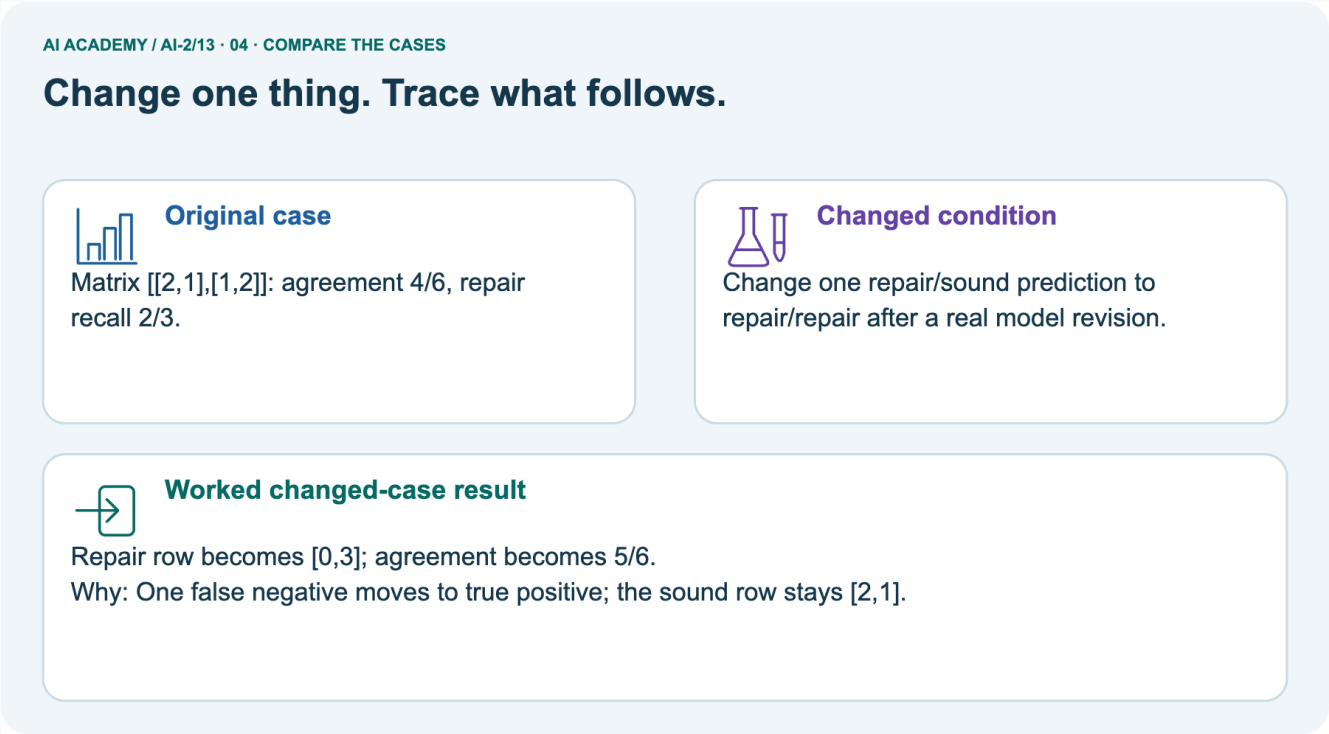
Follow the labelled picture with the matching explanation. Open the vector diagram

Reference rows, **prediction** columns. Reference sound: Predicted sound = 2, Predicted repair = 1. Reference repair: Predicted sound = 1, Predicted repair = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

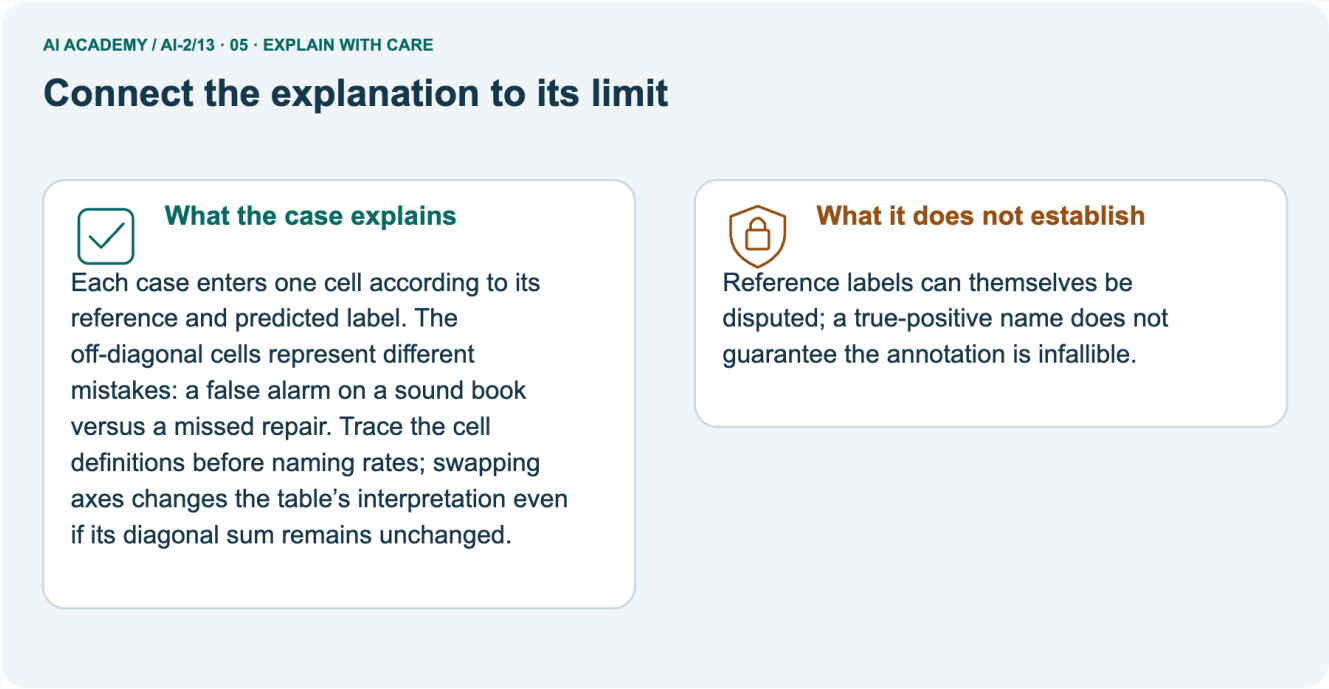
Original case: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Changed condition: Change one repair/sound **prediction** to repair/repair after a real model revision.

Expected result: Repair row becomes [0,3]; agreement becomes 5/6. Reason: One **false negative** moves to **true positive**; the sound row stays [2,1].

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

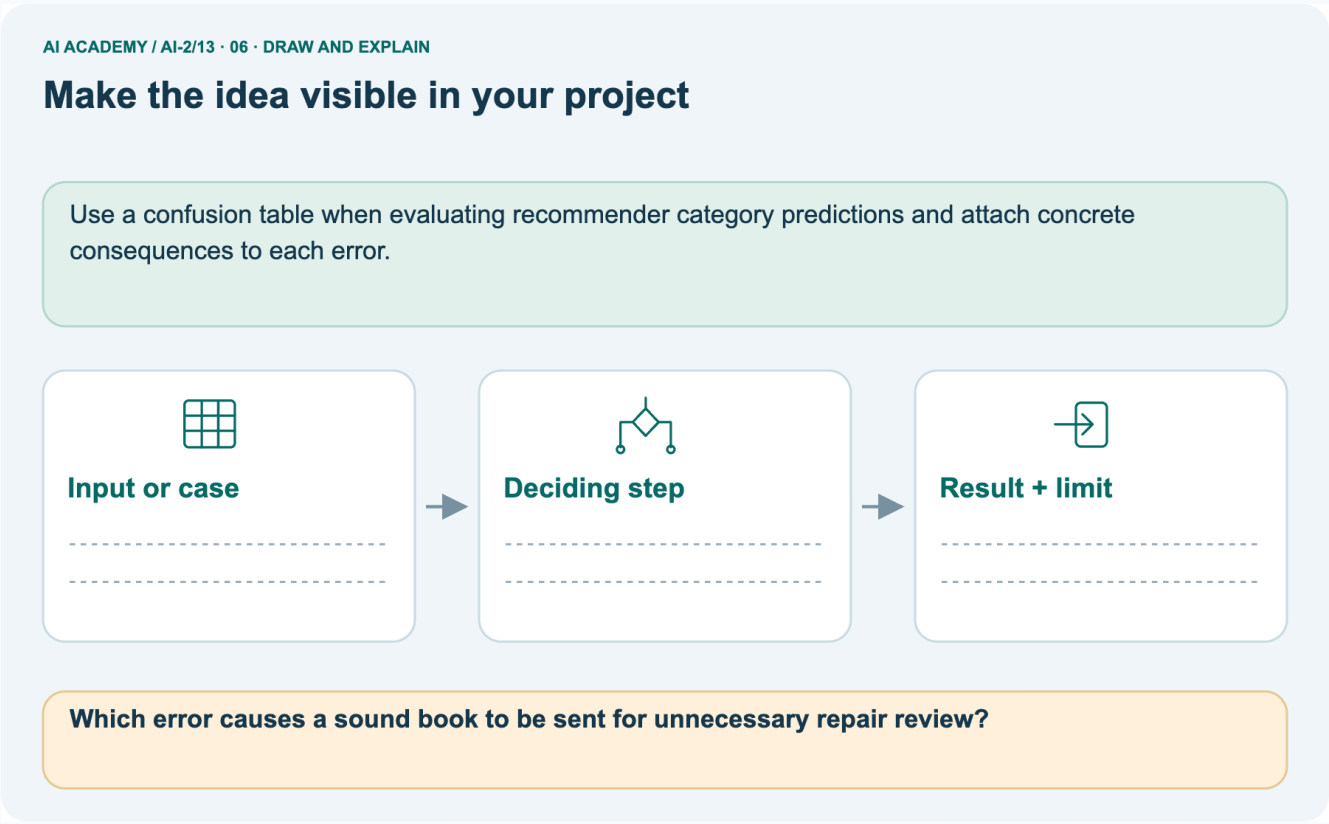
Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Limit: Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error. Draw your **input**, deciding step, result and limit.

Supported question: Which error causes a sound book to be sent for unnecessary repair review?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.